Application Layer Paradigms & Requirements

Coverage: Client-Server vs P2P Architectures, Transport Service Needs (Integrity, Throughput, Timing, Security), Socket Interface

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1. Application Paradigms

- Client-Server: Centralized server with fixed IP; clients initiate requests. Server upload bottlenecks at scale.
- Peer-to-Peer (P2P): Decentralized end hosts communicate directly; peers act as both clients and servers (self-scaling).

2. Transport Requirements

- Data Integrity: Loss-intolerant (TCP: Web, FTP, Email) vs Loss-tolerant (UDP: VoIP, Streaming).
- Timing & Throughput: Real-time interactive applications require tight delay bounds (<150ms).

■ KTU University High-Yield Exam Questions

- Q: Compare Client-Server and Peer-to-Peer network paradigms with real-world examples. (6 Marks)